

Contact

+21627954621 (Mobile)
mounir.mmahroug@gmail.com

www.linkedin.com/in/mounir-mahroug (LinkedIn)

Top Skills

Embedded C
Internet of Things (IoT)
Software Development

Languages

Anglais (Professional Working)
Arabic (Native or Bilingual)
French (Professional Working)

Certifications

Scrum Fundamentals Certified (SFC)
Advanced C Programming
Learning SCADA: Collect, Analyze, and Visualize Data for Industrial Automation
SolidWork(CSWA)
C Programming for Embedded Applications

MOUNIR MAHROUG

Embedded System Engineering
Ariana, Tunisia

Summary

As a freelance developer, I extend my skills to KiCad and EasyEDA for PCB design, and build full-stack IoT applications using Node.js, Next.js, and MongoDB, integrating embedded devices with secure cloud dashboards. I have experience in deployment, MQTT broker setup, and end-to-end IoT system delivery. Whether it's hardware design, firmware development, or cloud integration, I bring end-to-end expertise to deliver complete, reliable, and innovative solutions.

At SFM Telecom – SFM Technologies, I design and develop IoT solutions for a wide range of industrial applications. My experience covers embedded systems architecture, communication protocols, and development on platforms such as STM32 (Fxxx series), ESP32, and Raspberry Pi, as well as communication modules like SIM800L/SIM900/SIM7000G and Quectel EG95-EX. I code in C/C++ and Python, working with Modbus, I²C, SPI, MQTT, and Wi-Fi to deliver scalable, connected systems. I also use GitHub for version control and collaborative development.

I hold a degree in Mechatronics, Robotics, and Automation Engineering from ESPRIT (Ecole Supérieure Privée d'Ingénierie et de Technologies), where I learned to integrate mechanical, electrical, and software disciplines into cohesive system designs.

My certifications in SolidWorks, Scrum Fundamentals, and Embedded C Programming reinforce my multidisciplinary approach. I also specialize in PCB design, electronics hardware development, and computer-aided design (CAO). Passionate about embedded systems, I continually seek out new technologies and tools to stay at the forefront of innovation.

Experience

SERMA International

Electronic Engineer

September 2024 - Present (1 year 1 month)

Ariana, Tunisia

Freelance

Freelance

September 2024 - Present (1 year 1 month)

SFM Telecom-SFM Technologies

Embedded System Engineer || IoT solutions

May 2022 - August 2024 (2 years 4 months)

Tunis, Tunisia

SMDEi

Embedded Software Engineer

June 2021 - March 2022 (10 months)

TTeletronics Aerostanrew

Quality Control Technician

March 2018 - June 2021 (3 years 4 months)

Education

Ecole Supérieure Privée d'Ingénierie et de Technologies - ESPRIT

Excellent, Mechatronics, Robotics, and Automation Engineering · (September 2019 - May 2022)

Institute Superior of Science and Industrial Gabes

Electronic, Electrotechnical and Automatic · (September 2014 - June 2017)